

Adopted Levels, Gammas

$Q(\beta^-)=15753$ 10; $S(n)=859.8$ 87; $S(p)=2.344\times 10^4$ 60; $Q(\alpha)=-1.808\times 10^4$ 27

$Q(\beta^-), S(n), S(p), Q(\alpha)$: Deduced by the evaluators using mass excesses of 15529.5 71 for ^{35}Mg measured by 2025Ly01, and 8318 5 for ^{34}Mg ; a weighted average of 8323 7 (2019As04) and 8315 5 (2025Ly01); -224 7 for ^{35}Al , 31680 600 for ^{34}Na , and 31180 270 for ^{31}Ne from 2021Wa16. Values from 2021Wa16: $Q(\beta^-)=15860$ 270, $S(n)=750$ 270, $S(p)=23330$ 660, $Q(\alpha)=-17970$ 380.

$S(2n)=5576.0$ 76, $Q(\beta^-n)=10455.8$ 74, from mass excesses of 15529.5 71 for ^{35}Mg measured by 2025Ly01; 4962.9 27 for ^{33}Mg and -2997.6 21 for ^{34}Al from 2021Wa16. Values from 2021Wa16: $S(2n)=5470$ 270, $Q(\beta^-n)=10570$ 270.

$S(2p)=45070$ 660 (syst) (2021Wa16).

Isotope discovery (2012Th10): $\text{Ta}(^{48}\text{Ca}, X)$ projectile fragmentation at GANIL (1989Gu03, 1991Or01).

^{35}Mg production:

2012Kw02: ^{35}Mg produced by $\text{natNi}(^{40}\text{Ar}, X)$ at $E(^{40}\text{Ar})=140$ MeV/nucleon at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2011FuZZ: ^{35}Mg produced by $^9\text{Be}(^{48}\text{Ca}, X)$ fragmentation at $E(^{48}\text{Ca})=345$ MeV/nucleon at RIKEN. Measured thick target fragmentation, deduced production cross sections, and compared with EPAX-2.15 systematics.

2007Ts09: Analyzed fragmentation σ of n-rich Na, Mg isotopes (including ^{35}Mg) from $^9\text{Be}, ^{181}\text{Ta}(^{48}\text{Ca}, X)$. Used systematics based on average binding energy to extrapolate towards drip line, predicting σ for ^{40}Mg and discussing ^{39}Na . Compared different extrapolation models.

^{35}Mg decay measurements:

2013StZY: $^9\text{Be}(^{48}\text{Ca}, X)$ at RIKEN. Measured $T_{1/2}$.

1999YoZW: $^9\text{Be}(^{48}\text{Ca}, X)$ and $^{181}\text{Ta}(^{48}\text{Ca}, X)$ at RIKEN. Measured $T_{1/2}$ and $\% \beta^- n$.

^{35}Mg radius measurements:

2011Ka01: ^{35}Mg produced by $^9\text{Be}(^{48}\text{Ca}, X)$ fragmentation at GSI. Measured interaction cross sections with C and CH_2 targets at 900 MeV/nucleon. Deduced rms matter radii.

2006Kh08: ^{35}Mg produced by $^{181}\text{Ta}(^{48}\text{Ca}, X)$ fragmentation at $E(^{48}\text{Ca})=60.3$ MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

^{35}Mg mass measurements: 2025Ly01, 2007Ju03, 2001Sa72, 2000Sa21, 1991Or01.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J, π , mass, $T_{1/2}$, etc): 2023Ra22,

2021Ka07, 2020Mi15, 2016Ba59, 2016Sa46, 2016Sh05, 2015Sh21, 2014Ga13, 2014Wa14, 2013Ch31, 2013Li39, 2013Sh05,

2012Fo27, 2012Ho19, 2007Ha53, 2006Zh19, 2005Ch71, 2004Kh16, 1996Re10, 1991Pa19, 1991Pa21.

The level scheme is based on that of 2011Ga15 and 2017Mo26.

 ^{35}Mg LevelsCross Reference (XREF) Flags

- A $^{35}\text{Na} \beta^-$ decay (2.1 ms)
 B $^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$
 C $\text{C}(^{36}\text{Mg}, ^{35}\text{Mg}\gamma), (^{37}\text{Al}, ^{35}\text{Mg}\gamma)$

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0	$(3/2^-, 5/2^-)$	11.3 ms 6	BC	$\% \beta^- = 100$; $\% \beta^- n = 52$ 46; $\% \beta^- 2n = ?$ $\% \beta^- n$: from 1995ReZZ, 2008ReZZ. Other: 52 11 (1999YoZW, preliminary). Theoretical $\% \beta^- 0n = 29$, $\% \beta^- 1n = 66$, $\% \beta^- 2n = 5$ (2021Mi17). Theoretical $\% \beta^- 0n = 65$, $\% \beta^- 1n = 32$, $\% \beta^- 2n = 3$ (2019Mo01). J^π : $3/2^-$ from shell-model calculations with the SDPF-M and SDPF-M+2p $_{1/2}$ interactions (2017Mo26). Nearly degenerate 30-keV $3/2^-$ and $5/2^-$ g.s. from Monte Carlo shell-model calculations with the SDPF-M interaction (2011Ga15), and $3/2^-$ g.s. from shell-model calculations with the SDPF-U interaction (2011Ga15). $3/2^-$ from projection of the odd-neutron angular momentum along the symmetry axis and parity of the wave function (2019Mo01). Others: $3/2^+$ from antisymmetrized molecular dynamics (AMD) calculations with the Gogny DIS force (2017Mo26).

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Mg Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>XREF</u>	<u>Comments</u>
			$T_{1/2}$: 11.3 ms 5 (stat) 4 (syst) (2013StZY, implant- β correlation). Other: 72 ms 43 (2008ReZZ, 1995ReZZ) and ≈ 9 ms (1999YoZW, implant- β correlation, preliminary). Reduced strong absorption radius $r_0^2=1.64 \text{ fm}^2$ 15 from the energy-integrated σ of $\text{Si}(^{35}\text{Mg},\text{X})$ (2006Kh08). The rms matter radius $R_{\text{rms}}^{\text{m}}=3.40 \text{ fm}$ 24 from the interaction σ of $\text{C}, \text{CH}_2(^{35}\text{Mg},\text{X})$ (2011Ka01).
0+x		BC	E(level): $x \leq 80 \text{ keV}$ (2011Ga15 detection threshold); $x \leq 200 \text{ keV}$ (2017Mo26 detection threshold). J^π : Shell-model calculations with the SDPF-M+2p _{1/2} interaction predict a 1/2 ⁻ level at 141 keV (2017Mo26). Shell-model calculations with the SDPF-M interaction predict a 5/2 ⁻ level at 84 keV (2017Mo26). Monte Carlo shell-model calculations with the SDPF-M interaction predict a 3/2 ⁻ level at 30 keV (2011Ga15).
0+y?		C	E(level): $y \leq 200 \text{ keV}$ (2017Mo26 detection threshold). 2017Mo26 suggested a low-lying L=3 level from the observed L=3 component in the inclusive parallel momentum distribution.
206+x 8		C	J^π : 2017Mo26 states that based on the observed weak γ -ray intensity, this level is not the 1/2 ⁻ level at 141 keV predicted by shell-model calculations with the SDPF-M+2p _{1/2} interaction.
445+x 5	(3/2 ⁺ , 5/2 ⁺)	BC	J^π : L($^{36}\text{Mg}, ^{35}\text{Mg}$)=(2) from 0 ⁺ . 3/2 ⁺ from shell-model calculations with the SDPF-M+2p _{1/2} interaction (2017Mo26).
619+x 7	(1/2 ⁻ , 3/2 ⁻)	BC	J^π : L($^{36}\text{Mg}, ^{35}\text{Mg}$)=(1) from 0 ⁺ . 3/2 ⁻ from shell-model calculations with the SDPF-M+2p _{1/2} interaction (2017Mo26).
670+x 8		BC	

[†] From E γ data in ($^{38}\text{Si}, ^{35}\text{Mg}\gamma$) and ($^{36}\text{Mg}, ^{35}\text{Mg}\gamma$), ($^{37}\text{Al}, ^{35}\text{Mg}\gamma$).

 $\gamma(^{35}\text{Mg})$

<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_γ</u>	<u>I_γ</u>	<u>E_f</u>	<u>Comments</u>
206+x		206 8	100	0+x	E _γ : From 2017Mo26.
445+x	(3/2 ⁺ , 5/2 ⁺)	445 5	100	0+x	E _γ : weighted average of 443 7 (2017Mo26) and 446 5 (2011Ga15).
619+x	(1/2 ⁻ , 3/2 ⁻)	619 7	100	0+x	E _γ : weighted average of 616 8 (2017Mo26) and 621 7 (2011Ga15).
670+x		670 [†] 8	100	0+x	E _γ : From 2011Ga15, as this γ is not resolved from the 616 γ in 2017Mo26, but its presence is indicated in the fit of the spectrum. 2017Mo26 states that the origin of the 670 γ remains vague.

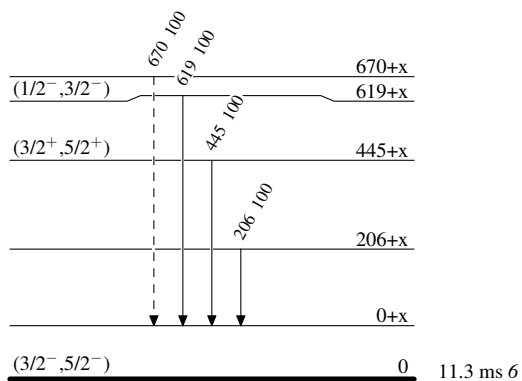
[†] Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{35}_{12}\text{Mg}_{23}$